Week 12 BA

2024 July07 Quiz-1

Comprehension for questions 1-3 (source Q5)

"Aqua Aqua (AA)" is a manufacturer of water purifiers. AA wants to understand the life of its RO filters. The life of a filter is measured in terms of "Litres of Water Purified (LWP)" before a warning light is raised for maintenance. A sample of 20 water purifiers were tested and the time (LWP) after which the warning light is raised for maintenance of the RO Filter was noted (data provided in Table-2). With this data, AA wants to understand the possible distribution for the life of the RO filter. To this end, it is proposed to bin the data in 4 groups: Low Life (LWP <= 25), Medium Life (25 < LWP <= 40), High Life (40 < LWP <= 80), and Excellent Life (LWP > 80). The following Φ values are given to you, where $\Phi(z)$ indicates the area under the curve (between "0" to "LWP") for a normal distribution with a mean of 60 and standard deviation of 24. $\Phi(-1.45) = 0.07$; $\Phi(-0.83) = 0.20$; $\Phi(+0.83) = 0.79$; $\Phi(+1.45) = 0.93$ Given this information, answer the subquestions

Sample-ID	RO Filter Life (in LWP)	Sample-ID	RO Filter Life (in LWP)
S-1	73	S-11	70
S-2	49	S-12	47
S-3	12	S-13	78
S-4	72	S-14	15
S-5	97	S-15	54
S-6	95	S-16	61
S-7	79	S-17	91
S-8	42	S-18	61
S-9	70	S-19	26
S-10	53	S-20	64

Table-2

1. [Source Q6; SA]
If the life of the RO Filter is expected to follow a uniform distribution, then what is the expected frequency in any given bin? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 21 to 23]

2. [Source Q7; SA]
If the life of the RO filter is expected to follow a uniform distribution, then what is the value of the computed test statistic? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 22]

3. [Source Q8; SA]
If the life of the RO filter is expected to follow a uniform distribution, then what is the degrees of freedom for the corresponding statistical test to verify the same? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 1]

2024 Oct27 Quiz-1

Comprehension for questions 4-6 (source Q3)

MSS is a firm that specializes in social media analytics. MSS has acquired data (in Table-1) on the past movie releases of two Heroes nick-named "Manda Kasayam (MK)" and "Youth Vodhavaka (YV)" in the Tamil Film industry. The data is on the Box-Office collections for the movies of MK and YV on different dates in January-1950. Then answer the given subquestions,with the presumption that the other necessary conditions required for the questions are met.

Date	Day	Box-Office collection for MK (in INR Million)	Box-Office Collection for YV (in INR Million)
01-Jan-1950	Sunday	11	10
02-Jan-1950	Monday	9	10
03-Jan-1950	Tuesday	15	9
04-Jan-1950	Wednesday	8	6
05-Jan-1950	Thursday	18	2
06-Jan-1950	Friday	16	10
07-Jan-1950	Saturday	19	7
08-Jan-1950	Sunday	8	6
09-Jan-1950	Monday	10	1
10-Jan-1950	Tuesday	19	3
11-Jan-1950	Wednesday	19	5
12-Jan-1950	Thursday	19	8
13-Jan-1950	Friday	15	8
14-Jan-1950	Saturday	17	5
15-Jan-1950	Sunday	15	2

Table-1

4. [Source Q9; SA]
There is a presumption that total box-office collection (of both actors) in Million INR on DIFFERENT DATES, follows a uniform distribution with a minimum value of 11 Million INR and maximum value of 30 Million INR. A statistical test is performed to check this presumption. The bins considered for the test are [11 to 15], [16 to 20], [21 to 25] and [26 to 30]. Then, for performing the statistical test, what is the expected frequency in any given bin? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 3.75]

5. [Source Q10; SA]
There is a presumption that total box-office collection (of both actors) in Million INR on DIFFERENT DATES, follows a uniform distribution with a minimum value of 11 Million INR and maximum value of 30 Million INR. A statistical test is performed to check this presumption. The bins considered for the test are [11 to 15], [16 to 20], [21 to 25] and [26 to 30]. Then, for performing the statistical test, what is the DOF? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 3]

6. [Source Q11; SA]
There is a presumption that total box-office collection (for both actors) in Million INR on DIFFERENT DATES, follows a uniform distribution with a minimum value of 11 Million INR and maximum value of 30 Million INR. A statistical test is performed to check this presumption. The bins considered for the test are [11 to 15], [16 to 20], [21 to 25] and [26 to 30]. Then, for performing the statistical test, what is the value of the computed test statistic? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 1.7 to 1.9]

2025 Apr13 ET

Comprehension for questions 7-10 (source Q23)

(The following is a purely imaginary scenario) Ms. Teddy, is the owner of a toy manufacturing company. The company produces its iconic "Teddy Bear" toys in a facility that operates for 8 hours a day, and 20 days in month. Ms. Teddy has recently completed the BA course and wants to see if her manufacturing facility is producing toys where the defects per ship follows a Poisson distribution. Accordingly, she collected data for the past month indicating the number of defects produced in a shift. This is provided in Table-1. Using this information answer the given sub-questions

Production Day in the Month	Number of Defective Teddy Bears Produced
Day-1	5
Day-2	0
Day-3	4
Day-4	0
Day-5	4
Day-6	4
Day-7	2
Day-8	5
Day-9	3
Day-10	3
Day-11	1
Day-12	5
Day-13	2
Day-14	1
Day-15	5
Day-16	5
Day-17	2
Day-18	1
Day-19	1
Day-20	1

Table-1

7. [Source Q24; SA]
How many bins will be present in the frequency table for the statistical test to be conducted? (Note: Enter an INTEGER answer)

[Source ans: 6]

8. [Source Q25; SA]
What is the value of the computed test statistic for the statistical test that is to be performed by Ms. Teddy to very her claim? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: 589.00 to 592.00]

9. [Source Q26; SA]
How many degrees of freedom is present for the test statistic that is to be conducted by Ms. Teddy? (Note: Enter an INTEGER answer)

[Source ans: 4]

10. [Source Q4; MSQ]

Given the Chisquare table in Figure-2, what is the conclusion from the test at a 95% significance level? (choose all that may be applicable)

ν \ α	.995	.990	.975	.950	.900	.500	.100	.050	.025	.010	.005
1	.00+	.00+	.00+	.00+	.02	.45	2.71	3.84	5.02	6.63	7.88
2	.01	.02	.05	.10	.21	1.39	4.61	5.99	7.38	9.21	10.60
3	.07	.11	.22	.35	.58	2.37	6.25	7.81	9.35	11.34	12.84
4	.21	.30	.48	.71	1.06	3.36	7.78	9.49	11.14	13.28	14.86
5	.41	.55	.83	1.15	1.61	4.35	9.24	11.07	12.83	15.09	16.75
6	.68	.87	1.24	1.64	2.20	5.35	10.65	12.59	14.45	16.81	18.55
7	.99	1.24	1.69	2.17	2.83	6.35	12.02	14.07	16.01	18.48	20.28
8	1.34	1.65	2.18	2.73	3.49	7.34	13.36	15.51	17.53	20.09	21.96
9	1.73	2.09	2.70	3.33	4.17	8.34	14.68	16.92	19.02	21.67	23.59
10	2.16	2.56	3.25	3.94	4.87	9.34	15.99	18.31	20.48	23.21	25.19
11	2.60	3.05	3.82	4.57	5.58	10.34	17.28	19.68	21.92	24.72	26.76
12	3.07	3.57	4.40	5.23	6.30	11.34	18.55	21.03	23.34	26.22	28.30
13	3.57	4.11	5.01	5.89	7.04	12.34	19.81	22.36	24.74	27.69	29.82
14	4.07	4.66	5.63	6.57	7.79	13.34	21.06	23.68	26.12	29.14	31.32
15	4.60	5.23	6.27	7.26	8.55	14.34	22.31	25.00	27.49	30.58	32.80

Figure-2

(a) REJECT the NULL HYPOTHESIS and conclude that the number of defectsFOLLOWS a poisson distribution

(b) DO NOT REJECT the NULL HYPOTHESIS and conclude that the number ofdefects FOLLOWS a poisson distribution

(c) REJECT the NULL HYPOTHESIS and conclude that the number of defects DOESNOT FOLLOW a poisson distribution

(d) DO NOT REJECT the NULL HYPOTHESIS and conclude that the number ofdefects DOES NOT FOLLOW a poisson distribution

(e) REJECT the ALTERNATIVE HYPOTHESIS and conclude that the number ofdefects FOLLOWS a poisson distribution

(f) DO NOT REJECT the ALTERNATIVE HYPOTHESIS and conclude that the numberof defects FOLLOWS a poisson distribution

(g) REJECT the ALTERNATIVE HYPOTHESIS and conclude that the number ofdefects DOES NOT FOLLOW a poisson distribution

(h) DO NOT REJECT the ALTERNATIVE HYPOTHESIS and conclude that the numberof defects DOES NOT FOLLOW a poisson distribution

[Source ans: c]

2025 Oct26 Quiz-1

Comprehension for questions 11-14 (source Q12)

(The following is a partially imaginary scenario) The BA discourse is often cluttered with queries / requests related to solutions for Past Year Question Papers (PYQP). The course instructor, Dr. Milo, had been informing that students need to put in effort to find solutions by searching the discourse. However, the results seem to be disappointing. Therefore, Dr. Milo wanted to do a few basic analytics on discourse behaviour of students. Past student data was collected for the analysis. Past data was collected for the BA terms in Jan-2022, May-2022, Sep-2022, Jan-2023, May-2023 and Sep-2023. For each term, 150 randomly selected unique student IDs were sampled. Of the 150 student IDs, 75 were "Male" and the remaining were "Female". The learners in the discourse have either "followed the instructions" or "not followed the instructions" or "have not accessed the discourse at all" (these three are referred to as access categories). Specifically, for the sample data in the different terms (in sequence of earliest term to later term i.e. Jan-2022 to Sep-2023) it was observed that 1. For the Male sample: 30, 20, 35, 35, 45 and 35 had not followed the instructions respectively in each term (i.e 30 male learners had not followed the instructions in Jan-2022 term and so on) 2. For the Male sample: 20, 10, 15, 15, 15 and 10 had not accessed discourse at all respectively in each term (i.e 20 male learners had not accessed discourse at all in Jan-2022 term and so on) 3. For the female sample: 15, 22, 33, 25, 20 and 20 had followed the instructions respectively in each term (i.e 15 female learners had followed the instructions in Jan-2022 term and so on) 4. For the female sample: 30, 20, 25, 25, 25 and 10 had not accessed discourse at all respectively in each term (i.e 30 female learners had not accessed discourse at all in Jan-2022 term and so on) Moreover, Dr. Milo assumes that all learners in the "followed the instructions" category are Uniformly distributed between 4 and 75. A total of 6 bins of equal area are created for performing the appropriate statistical test. Each term is a sample point. Given this information, answer the sub questions.

11. [Source Q13; SA]
What is the probability that a specific term will fall in Bin-4 when performing the hypothesis test to check if Dr. Milo's assumption that all learners in the "followed the instructions" category are Uniformly distributed? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "0.235", then enter it as "0.24")

[Source ans: 0.16 to 0.18]

12. [Source Q14; SA]
What will be the expected frequency in Bin-2 when performing the hypothesis test to check if Dr. Milo's assumption that all learners in the "followed the instructions" category are Uniformly distributed? (Note: Enter the answer rounded to two decimal places . For example, if the answer is "1.235", then enter it as "1.24")

[Source ans: 1]

13. [Source Q15; SA]
What is the value of the computed test statistic which will be used to check if Dr. Milo's assumption that all learners in the "followed the instructions" category are Uniformly distributed? (Note: Enter the answer rounded to two decimal places . For example, if the answer is "1.235", then enter it as "1.24")

[Source ans: 13.20 to 13.90]

14. [Source Q16; SA]
How many degrees of freedom is present for the hypothesis test used to check if Dr. Milo's assumption that all learners in the "followed the instructions" category are Uniformly distributed?

[Source ans: 5]

2026 Mar15 Quiz-1

Comprehension for questions 15-18 (source Q16)

Ms. Teddy, is the owner of a toy manufacturing company. The company produces its iconic "Teddy Bear" toys in a facility that operates for 8 hours a day, and 20 days in month. Ms. Teddy has recently completed the BA course and wants to see if her manufacturing facility is producing toys where the defects follows a Poisson distribution. Accordingly, she collected 500 samples from the production and examined them for defects. This information is provided in Table-1. Ms. Teddy observed that her production facility never produced a product that had more than 4 defects.

Number of Defects	Number of Samples observed
1	120
2	80
3	150
4	50

Table-1: Data on Teddy Bears that were collected Based on the above data, answer the given subquestions.

15. [Source Q17; SA]
How many teddy bears will Ms. Teddy expect to see which will have "4 defects"? (Note: Input your answer rounded to two decimal places. For example, if your answer is "1.235" then enter the answer as "1.24")

[Source ans: 38.1 to 38.90]

16. [Source Q18; SA]
How many degrees of freedom is/are present for the statistical test to be performed by Ms. Teddy?

[Source ans: 4]

17. [Source Q19; SA]
What is the value of the computed test statistic for the statistical test to be performed by Ms. Teddy? (Note: Input your answer rounded to two decimal places. For example, if your answer is "1.235" then enter the answer as "1.24")

[Source ans: 81.00 to 83.00]

18. [Source Q20; MSQ]

Given Figure-1, which of the following statements is/are true (choose all that is/ are applicable)

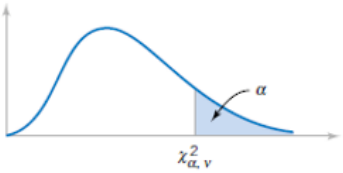


Table III Percentage Points $\chi^2_{\alpha,\nu}$ of the Chi-Squared Distribution

ν \ α	.995	.990	.975	.950	.900	.500	.100	.050	.025	.010	.005
1	.00+	.00+	.00+	.00+	.02	.45	2.71	3.84	5.02	6.63	7.88
2	.01	.02	.05	.10	.21	1.39	4.61	5.99	7.38	9.21	10.60
3	.07	.11	.22	.35	.58	2.37	6.25	7.81	9.35	11.34	12.84
4	.21	.30	.48	.71	1.06	3.36	7.78	9.49	11.14	13.28	14.86
5	.41	.55	.83	1.15	1.61	4.35	9.24	11.07	12.83	15.09	16.75
6	.68	.87	1.24	1.64	2.20	5.35	10.65	12.59	14.45	16.81	18.55
7	.99	1.24	1.69	2.17	2.83	6.35	12.02	14.07	16.01	18.48	20.28
8	1.34	1.65	2.18	2.73	3.49	7.34	13.36	15.51	17.53	20.09	21.96
9	1.73	2.09	2.70	3.33	4.17	8.34	14.68	16.92	19.02	21.67	23.59
10	2.16	2.56	3.25	3.94	4.87	9.34	15.99	18.31	20.48	23.21	25.19
11	2.60	3.05	3.82	4.57	5.58	10.34	17.28	19.68	21.92	24.72	26.76
12	3.07	3.57	4.40	5.23	6.30	11.34	18.55	21.03	23.34	26.22	28.30
13	3.57	4.11	5.01	5.89	7.04	12.34	19.81	22.36	24.74	27.69	29.82
14	4.07	4.66	5.63	6.57	7.79	13.34	21.06	23.68	26.12	29.14	31.32
15	4.60	5.23	6.27	7.26	8.55	14.34	22.31	25.00	27.49	30.58	32.80
16	5.14	5.81	6.91	7.96	9.31	15.34	23.54	26.30	28.85	32.00	34.27
17	5.70	6.41	7.56	8.67	10.09	16.34	24.77	27.59	30.19	33.41	35.72
18	6.26	7.01	8.23	9.39	10.87	17.34	25.99	28.87	31.53	34.81	37.16
19	6.84	7.63	8.91	10.12	11.65	18.34	27.20	30.14	32.85	36.19	38.58
20	7.43	8.26	9.59	10.85	12.44	19.34	28.41	31.41	34.17	37.57	40.00
21	8.03	8.90	10.28	11.59	13.24	20.34	29.62	32.67	35.48	38.93	41.40
22	8.64	9.54	10.98	12.34	14.04	21.34	30.81	33.92	36.78	40.29	42.80
23	9.26	10.20	11.69	13.09	14.85	22.34	32.01	35.17	38.08	41.64	44.18
24	9.89	10.86	12.40	13.85	15.66	23.34	33.20	36.42	39.36	42.98	45.56
25	10.52	11.52	13.12	14.61	16.47	24.34	34.28	37.65	40.65	44.31	46.93
26	11.16	12.20	13.84	15.38	17.29	25.34	35.56	38.89	41.92	45.64	48.29
27	11.81	12.88	14.57	16.15	18.11	26.34	36.74	40.11	43.19	46.96	49.65
28	12.46	13.57	15.31	16.93	18.94	27.34	37.92	41.34	44.46	48.28	50.99
29	13.12	14.26	16.05	17.71	19.77	28.34	39.09	42.56	45.72	49.59	52.34
30	13.79	14.95	16.79	18.49	20.60	29.34	40.26	43.77	46.98	50.89	53.67
40	20.71	22.16	24.43	26.51	29.05	39.34	51.81	55.76	59.34	63.69	66.77
50	27.99	29.71	32.36	34.76	37.69	49.33	63.17	67.50	71.42	76.15	79.49
60	35.53	37.48	40.48	43.19	46.46	59.33	74.40	79.08	83.30	88.38	91.95
70	43.28	45.44	48.76	51.74	55.33	69.33	85.53	90.53	95.02	100.42	104.22
80	51.17	53.54	57.15	60.39	64.28	79.33	96.58	101.88	106.63	112.33	116.32
90	59.20	61.75	65.65	69.13	73.29	89.33	107.57	113.14	118.14	124.12	128.30

Figure-1

(a) The p-value for the Test is anywhere between 5% and 2.5%

(b) At a 5% confidence level, the Null Hypothesis is Not Rejected

(c) At a 5% confidence level, the Null Hypothesis is Rejected

(d) At a 5% significance level, the Null Hypothesis is Not Rejected

(e) At a 5% significance level, the Null Hypothesis is Rejected

(f) At a 2.5% significance level, the Tabulated Chi-square value for the test is 12.83

(g) None of these

[Source ans: c, e]

2026 May06 ET

Comprehension for questions 19-23 (source Q41)

The following scenario is purely hypothetical The daily sales of choco-chip ice-cream cones in an ice-cream parlour is assumed to follow a uniform distribution with a minimum sales of 10 and a maximum sales of 23. To check this assumption, the sales of choco-chip ice-cream cones at the ice-cream parlour for the past 14 days was collected (Table-4). The frequency table is developed with 7 bins. With this data, answer the given subquestions

Day	The actual number of choco-chip ice-cream cones sold	Day	The actual number of choco-chip ice-cream cones sold
Day-1	17	Day-8	10
Day-2	19	Day-9	14
Day-3	22	Day-10	12
Day-4	13	Day-11	23
Day-5	12	Day-12	22
Day-6	21	Day-13	11
Day-7	18	Day-14	22

Table-4

19. [Source Q42; SA]
What is the expected frequency in Day-2? (Note: Enter your answer in decimal values rounded to two decimal places. For example, if your answer is "1.235", then enter the answer as "1.24")

[Source ans: 1.90 to 2.10]

20. [Source Q43; SA]
What is the value for the computed test statistic? (Note: Enter your answer in decimal values rounded to two decimal places. For example, if your answer is "1.235", then enter the answer as "1.24")

[Source ans: 4]

21. [Source Q44; SA]
How many degrees of freedom does that Chi-Square Goodness of Fit Test statistic have? (Note: Enter your answer in decimal values rounded to two decimal places. For example, if your answer is "1.235", then enter the answer as "1.24")

[Source ans: 6]

22. [Source Q45; MSQ]
Which of the following is the correct null hypothesis for the performed hypothesis test? (choose all that is/ are applicable)

(a) The sales of choco-chip ice-creams at the ice-cream parlour is uniformly distributed

(b) The daily-sales of choco-chip ice-creams at the ice-cream parlour is uniformly distributed

(c) The sales of choco-chip ice-creams at the ice-cream parlour is NOT uniformly distributed

(d) The daily-sales of choco-chip ice-creams at the ice-cream parlour is NOT uniformly distributed

[Source ans: b]

23. [Source Q46; MSQ]

Given the chi-square table below, what is your conclusion. Choose all that is/are applicable

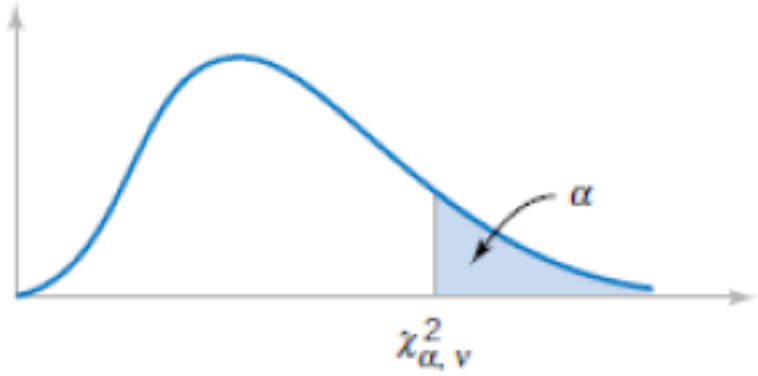


Table III Percentage Points $\chi^2_{\alpha,\nu}$ of the Chi-Squared Distribution

ν \ α	.995	.990	.975	.950	.900	.500	.100	.050	.025	.010	.005
1	.00+	.00+	.00+	.00+	.02	.45	2.71	3.84	5.02	6.63	7.88
2	.01	.02	.05	.10	.21	1.39	4.61	5.99	7.38	9.21	10.60
3	.07	.11	.22	.35	.58	2.37	6.25	7.81	9.35	11.34	12.84
4	.21	.30	.48	.71	1.06	3.36	7.78	9.49	11.14	13.28	14.86
5	.41	.55	.83	1.15	1.61	4.35	9.24	11.07	12.83	15.09	16.75
6	.68	.87	1.24	1.64	2.20	5.35	10.65	12.59	14.45	16.81	18.55
7	.99	1.24	1.69	2.17	2.83	6.35	12.02	14.07	16.01	18.48	20.28
8	1.34	1.65	2.18	2.73	3.49	7.34	13.36	15.51	17.53	20.09	21.96
9	1.73	2.09	2.70	3.33	4.17	8.34	14.68	16.92	19.02	21.67	23.59
10	2.16	2.56	3.25	3.94	4.87	9.34	15.99	18.31	20.48	23.21	25.19
11	2.60	3.05	3.82	4.57	5.58	10.34	17.28	19.68	21.92	24.72	26.76
12	3.07	3.57	4.40	5.23	6.30	11.34	18.55	21.03	23.34	26.22	28.30
13	3.57	4.11	5.01	5.89	7.04	12.34	19.81	22.36	24.74	27.69	29.82
14	4.07	4.66	5.63	6.57	7.79	13.34	21.06	23.68	26.12	29.14	31.32
15	4.60	5.23	6.27	7.26	8.55	14.34	22.31	25.00	27.49	30.58	32.80
16	5.14	5.81	6.91	7.96	9.31	15.34	23.54	26.30	28.85	32.00	34.27
17	5.70	6.41	7.56	8.67	10.09	16.34	24.77	27.59	30.19	33.41	35.72
18	6.26	7.01	8.23	9.39	10.87	17.34	25.99	28.87	31.53	34.81	37.16
19	6.84	7.63	8.91	10.12	11.65	18.34	27.20	30.14	32.85	36.19	38.58
20	7.43	8.26	9.59	10.85	12.44	19.34	28.41	31.41	34.17	37.57	40.00
21	8.03	8.90	10.28	11.59	13.24	20.34	29.62	32.67	35.48	38.93	41.40
22	8.64	9.54	10.98	12.34	14.04	21.34	30.81	33.92	36.78	40.29	42.80
23	9.26	10.20	11.69	13.09	14.85	22.34	32.01	35.17	38.08	41.64	44.18
24	9.89	10.86	12.40	13.85	15.66	23.34	33.20	36.42	39.36	42.98	45.56
25	10.52	11.52	13.12	14.61	16.47	24.34	34.28	37.65	40.65	44.31	46.93
26	11.16	12.20	13.84	15.38	17.29	25.34	35.56	38.89	41.92	45.64	48.29
27	11.81	12.88	14.57	16.15	18.11	26.34	36.74	40.11	43.19	46.96	49.65
28	12.46	13.57	15.31	16.93	18.94	27.34	37.92	41.34	44.46	48.28	50.99
29	13.12	14.26	16.05	17.71	19.77	28.34	39.09	42.56	45.72	49.59	52.34
30	13.79	14.95	16.79	18.49	20.60	29.34	40.26	43.77	46.98	50.89	53.67
40	20.71	22.16	24.43	26.51	29.05	39.34	51.81	55.76	59.34	63.69	66.77
50	27.99	29.71	32.36	34.76	37.69	49.33	63.17	67.50	71.42	76.15	79.49
60	35.53	37.48	40.48	43.19	46.46	59.33	74.40	79.08	83.30	88.38	91.95
70	43.28	45.44	48.76	51.74	55.33	69.33	85.53	90.53	95.02	100.42	104.22
80	51.17	53.54	57.15	60.39	64.28	79.33	96.58	101.88	106.63	112.33	116.32
90	59.20	61.75	65.65	69.13	73.29	89.33	107.57	113.14	118.14	124.12	128.30

(a) Based on the computed test statistic, the NULL hypothesis is never rejected

(b) Based on the computed test statistic, the NULL hypothesis is always rejected

(c) Based on the computed test statistic, the NULL hypothesis is rejected for all significance levels greater than 5%

(d) Based on the computed test statistic, the NULL hypothesis is NOT rejected for all significance levels greater than 50%

(e) Based on the computed test statistic, the NULL hypothesis is rejected for all confidence levels less than 95%

(f) Based on the computed test statistic, the NULL hypothesis is rejected for all confidence levels greater than 95%

(g) None of these

[Source ans: d, e]

2026 Jul16 Quiz-1

Comprehension for questions 24-27 (source Q6)

(The following comprehension is purely imaginary) The BA team is known for their clumsy QP setting. Hence, after every quiz the enrolled learners always raise concerns. These concerns can be grouped into five categories: "Category-1: Not a valid concern- No Correction", "Category-2: Valid Concern-Minor Correction", "Category-3: Valid Concern-Major Corrections", "Catgeory-4: Valid Concern-Open Marks" and "Catgeory-5: Those who claim marks for their misconception". From the past six terms (Jan-20 to Sep-21), data on the number of requests in each category is captured and presented in Table-2 below. Given this information, answer the given sub-questions

	Jan-20	May-20	Sep-20	Jan-21	May-21	Sep-21	***Row Total***
Category-1	1	2	1	2	2	2	*10*
Category-2	5	2	2	0	2	4	*15*
Category-3	2	6	6	5	5	1	*25*
Category-4	1	2	1	2	2	2	*10*
Category-5	10	12	4	3	3	8	*40*
Column Total	*19*	*24*	*14*	*12*	*14*	*17*	

Table-2: Category-Term Count Data

24. [Source Q10; SA]
Say, the goal is to find the distribution of the counts in "Category-4". If each year is taken as a data point and the assumption is that the counts in Catgeory-4 are distributed as a Poisson, then how many degrees of freedom is present for the computed test statistic (Hint: The last bin in the frequency table is based on the provided sample)?

[Source ans: 1]

25. [Source Q11; SA]
Say, the goal is to find the distribution of the counts in "Category-4". If each year is taken as a data point and the assumption is that the counts in Catgeory-4 are distributed as a Poisson, then what is the expected frequency for the count of "1"? (Note: Enter your answer in decimals rounded to two decimal places. For example, if your answer is "0.245", enter it as "0.25")

[Source ans: 1.50 to 2.10]

26. [Source Q12; SA]
Say, the goal is to find the distribution of the counts in "Category-4". If each year is taken as a data point and the assumption is that the counts in Catgeory-4 are distributed as a Poisson, then what is value for the computed test statistic? (Hint: The last bin in the frequency table is based on the provided sample)(Note: Enter your answer in decimals rounded to two decimal places. For example, if your answer is "0.245", enter it as "0.25")

[Source ans: 1.2 to 1.6]

27. [Source Q13; MSQ]
Figure-1 (shared reference table from source Q9):

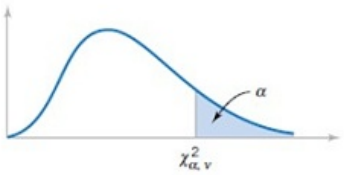


Table III Percentage Points $\chi^2_{\alpha,\nu}$ of the Chi-Squared Distribution

ν \ α	.995	.990	.975	.950	.900	.500	.100	.050	.025	.010	.005
1	.00+	.00+	.00+	.00+	.02	.45	2.71	3.84	5.02	6.63	7.88
2	.01	.02	.05	.10	.21	1.39	4.61	5.99	7.38	9.21	10.60
3	.07	.11	.22	.35	.58	2.37	6.25	7.81	9.35	11.34	12.84
4	.21	.30	.48	.71	1.06	3.36	7.78	9.49	11.14	13.28	14.86
5	.41	.55	.83	1.15	1.61	4.35	9.24	11.07	12.83	15.09	16.75
6	.68	.87	1.24	1.64	2.20	5.35	10.65	12.59	14.45	16.81	18.55
7	.99	1.24	1.69	2.17	2.83	6.35	12.02	14.07	16.01	18.48	20.28
8	1.34	1.65	2.18	2.73	3.49	7.34	13.36	15.51	17.53	20.09	21.96
9	1.73	2.09	2.70	3.33	4.17	8.34	14.68	16.92	19.02	21.67	23.59
10	2.16	2.56	3.25	3.94	4.87	9.34	15.99	18.31	20.48	23.21	25.19
11	2.60	3.05	3.82	4.57	5.58	10.34	17.28	19.68	21.92	24.72	26.76
12	3.07	3.57	4.40	5.23	6.30	11.34	18.55	21.03	23.34	26.22	28.30
13	3.57	4.11	5.01	5.89	7.04	12.34	19.81	22.36	24.74	27.69	29.82
14	4.07	4.66	5.63	6.57	7.79	13.34	21.06	23.68	26.12	29.14	31.32
15	4.60	5.23	6.27	7.26	8.55	14.34	22.31	25.00	27.49	30.58	32.80
16	5.14	5.81	6.91	7.96	9.31	15.34	23.54	26.30	28.85	32.00	34.27
17	5.70	6.41	7.56	8.67	10.09	16.34	24.77	27.59	30.19	33.41	35.72
18	6.26	7.01	8.23	9.39	10.87	17.34	25.99	28.87	31.53	34.81	37.16
19	6.84	7.63	8.91	10.12	11.65	18.34	27.20	30.14	32.85	36.19	38.58
20	7.43	8.26	9.59	10.85	12.44	19.34	28.41	31.41	34.17	37.57	40.00
21	8.03	8.90	10.28	11.59	13.24	20.34	29.62	32.67	35.48	38.93	41.40
22	8.64	9.54	10.98	12.34	14.04	21.34	30.81	33.92	36.78	40.29	42.80
23	9.26	10.20	11.69	13.09	14.85	22.34	32.01	35.17	38.08	41.64	44.18
24	9.89	10.86	12.40	13.85	15.66	23.34	33.20	36.42	39.36	42.98	45.56
25	10.52	11.52	13.12	14.61	16.47	24.34	34.28	37.65	40.65	44.31	46.93
26	11.16	12.20	13.84	15.38	17.29	25.34	35.56	38.89	41.92	45.64	48.29
27	11.81	12.88	14.57	16.15	18.11	26.34	36.74	40.11	43.19	46.96	49.65
28	12.46	13.57	15.31	16.93	18.94	27.34	37.92	41.34	44.46	48.28	50.99
29	13.12	14.26	16.05	17.71	19.77	28.34	39.09	42.56	45.72	49.59	52.34
30	13.79	14.95	16.79	18.49	20.60	29.34	40.26	43.77	46.98	50.89	53.67

Say, the goal is to find the distribution of the counts in "Category-4". If each year is taken as a data point and the assumption is that the counts in Catgeory-4 are distributed as a Poisson, then using Figure-1, based on the computed test statistic which of the following are TRUE (choose all that is/are applicable)

(a) Reject the Null Hypothesis

(b) Do Not Reject the Null Hypothesis

(c) The p-value for the test is between 0.5 to 0.1 and more towards 0.1

(d) Insufficient information. Cannot make a conclusion

[Source ans: c, d]